Gymnastiar Al Khoarizmy

Data Science Graduate
Institut Teknologi Sumatera

+62 878-6081-1076
gymnastiar12345678@gmail.com
GitHub Profile
LinkedIn Profile



Summary

Data Science graduate from Institut Teknologi Sumatera with practical experience in data engineering, SQL analytics, ETL pipeline development, and applied machine learning. Experienced in designing automated data collection systems, data validation pipelines, and interactive BI dashboards across public sector and operational datasets. Proficient in Python, SQL, DuckDB, Tableau, and statistical model evaluation.

Education

•Bachelor of Science in Data Science *2022 – 2026*
Institut Teknologi Sumatera, Lampung GPA: 3.81/4.00

Professional Experience

•Investment Data Analyst Intern *Jun 2025 – Sep 2025*
Ministry of Investment and Downstream Industry / BKPM Jakarta, Indonesia

- Developed an Excel VBA automation tool to standardize and validate quarterly investment compliance records across multi-company project databases.
- Built a predictive machine learning model to estimate investment project transition timelines from construction to commercial production phase.
- Performed exploratory data analysis, feature engineering, and cross-validation on provincial investment datasets to identify reporting delay factors.
- Presented analytical findings and project duration estimates directly to department directors to support investment monitoring workflows.

•Data Science & Programming Practicum Assistant *Jul 2024 – Dec 2025*
Institut Teknologi Sumatera Lampung, Indonesia

- Mentored undergraduate students in Python, R, SQL, data mining, statistical modeling, hypothesis testing, ANOVA, and machine learning evaluation.
- Guided students through code debugging, data preprocessing pipelines, and statistical interpretation during weekly hands-on laboratory sessions.
- Evaluated weekly programming assignments and provided technical feedback on data cleaning standards and model validation methodology.

Featured Projects

•Automated Public Procurement Data Pipeline and BI Dashboard *Personal Project*
Python, DuckDB, Google Sheets API, Tableau Public, Diagrams-as-Code Article / Tableau / GitHub

- Built an automated end-to-end data pipeline to ingest, clean, and analyze public procurement records across multiple government ministries and agencies.
- Implemented parallel workers for data extraction, schema normalization, date parsing, and rule-based record deduplication.
- Automated cloud data synchronization to Google Sheets and designed an interactive Tableau Public dashboard with scheduled auto-refresh.
- Executed analytical SQL queries using DuckDB to evaluate budget efficiency gaps, procurement method distributions, and vendor concentration patterns.

•Zero-Shot Coastal Waste Detection and Material Classification *Final Project / Thesis*
Grounding DINO, CLIP, Computer Vision, Model Evaluation

- Developed a zero-shot object detection and classification pipeline for coastal waste identification across beach imagery in Lampung Province.
- Evaluated prompt engineering scenarios and model performance using IoU, precision, recall, and F1-score across multiple material classes.
- Analyzed error distributions, false positives, and class imbalance to assess model behavior and improve data annotation quality.

•**Fine-Tuning Qwen3-TTS for Indonesian Voice Synthesis** *Personal Project*
Qwen3-TTS, PyTorch, Speech Synthesis, Google Colab Article
– Fine-tuned an open-weight speech synthesis model (Qwen3-TTS) for Bahasa Indonesia using public speech corpora on GPU environments.
– Evaluated phoneme pronunciation naturalness, acoustic stability, and speaker timbre consistency across generated Indonesian speech samples.
– Published a reproducible fine-tuning guide and open-source implementation to support local language speech technology development.

•**End-to-End Spoken Language Understanding for Retail Transactions** *Course Project*
CNN-BiLSTM, PyTorch, Deep Learning, Speech Processing GitHub
– Built an Indonesian spoken language understanding pipeline for retail transaction voice logging using sequence modeling architectures.
– Developed audio preprocessing, intent classification, and slot-filling modules to extract product names, quantities, and transaction details from speech.
– Evaluated model performance across diverse acoustic environments to ensure classification accuracy and reproducible experimental workflows.

•**Plant Disease Diagnosis System using MLOps and Active Learning** *INNOVEST 2026 – 3rd Winner*
Machine Learning Operations, Active Learning, Data Labeling Workflow
– Designed an active learning image classification pipeline for agricultural crop disease diagnosis to optimize data labeling efficiency.
– Implemented uncertainty-based sampling to selectively route ambiguous field samples to domain experts for iterative model retraining.
– Awarded 3rd Place at INNOVEST 2026 National Competition for system architecture and data annotation efficiency design.

•**Seasonal Forecasting of Ferry Passenger Demand** *Research Project*
Time Series Forecasting, SARIMA, Transportation Analytics Article
– Conducted time series analysis and seasonal demand forecasting on historical passenger transit data at Bakauheni Port, Indonesia.
– Implemented SARIMA modeling with stationarity transformations and residual diagnostics to forecast seasonal passenger traffic patterns.
– Published research methodology and operational demand forecasting findings in the peer-reviewed IJECS journal.

Publications & Achievements

•**Peer-Reviewed Journal Publication** – IJECS (Vol. 4, No. 2, 2025) *2025*
– Published research paper: "Forecasting Ferry Passengers at Bakauheni Port using Seasonal Autoregressive Integrated Moving Average (SARIMA)."

•**3rd Winner** – INNOVEST 2026 National Competition, Politeknik Negeri Jember *2026*
– Awarded 3rd Winner for the project "Implementation of a Plant Disease Diagnosis System with Machine Learning Operations (MLOps) and Active Learning."

Technical Skills

Programming: Python, R, SQL, Bash, Excel VBA

Data Engineering & BI: ETL Pipelines, DuckDB, Google Sheets API, Tableau Public, Looker Studio, Web Scraping

Data Science & ML: Machine Learning, Deep Learning, Time Series Forecasting, Computer Vision, Model Evaluation, Data Validation

Developer Tools: Git, GitHub, Docker, Streamlit, Jupyter Notebook, LaTeX

Languages: Indonesian (Native), English (Professional)

Certifications

•**Python Project for Data Science** – IBM Coursera
•**Python for Data Science, AI & Development** – IBM Coursera
•**Introduction to Financial Accounting** – University of Pennsylvania, Coursera